

P2_reference_diagnosis

2026.07.27 - How do we know run-3 P2 has a wild-type reference problem?

Every number below is regenerated by a committed script reading the real result files: `between_day_variance.py`, `p2_reference_diagnosis.py`, `verify_plate_pairing.py`, `replication_structure.py`, `ladder_feasibility.py`.

The observation

Fitness is scored as `median(norm | strain) / median(norm | BY4741)`, so the wild-type median is the denominator of **every** value on a plate. Comparing the wild type against that same plate's own mutants, in raw pixels (normalization-independent):

plate	WT median px	mutant median px	WT / mutant	WT_CV
run2 P2 t44	1983	1584	1.252	0.111
run2 P2 t50	233	199	1.171	0.083
run3 P1	3568	2773	1.287	0.066
run3 P2	2412	2703	0.892	0.144
run3 P3	3531	2665	1.325	0.079

Four plates sit at 1.17-1.33. P2 alone is below 1. (The t50 pixel counts are small because that capture is at a different resolution; the *ratio* is scale-free, which is why the comparison is made on a ratio.)

Nine things it is NOT

1. Not a plate-wide growth, imaging or exposure problem. The mutants are the same size on all three run-3 plates – **2773 / 2703 / 2664 px, within 4%**. Only the wild type moves: 3568 / **2412** / 3531, so P2's WT is **32% smaller** while everything else is unchanged. Any plate-level cause (agar batch, incubation slot, exposure, focus) would move both.

2. Not a mis-mapped picklist. The image-to-picklist pairing was re-derived from content across all 3 images x 3 picklists x 4 orientations, not taken from filenames:

Blank wells landing on empty wells (max 6):

image picklist	Plate1	Plate2	Plate3
P1	6	0	0
P2	0	6	1
P3	0	1	6

Kruskal-Wallis H across strain groups:

image picklist	Plate1	Plate2	Plate3
P1	97.1	14.1	13.9
P2	13.7	146.3	14.7
P3	19.4	11.2	125.9

Every off-diagonal cell sits at the null (~12 for 13 groups). Total H is 369.3 for the filename assignment against 179.6 for the next-best permutation. Critically, **P2 has the HIGHEST H of the three plates** – a mis-mapped plate would collapse H toward the null, not maximize it, because every “strain” group would become a mixture of all genotypes.

3. Not a wrong orientation. All plates resolve to **identity** (A1 top-left), which matches the imaging protocol (A1 top-left, agar side up) and is independently consistent with the chamfered corners sitting at the bottom of every image.

4. Not a dispense failure or a drained source well. From the Echo’s own transfer report: BY4741 is drawn from source well **A1 on all three plates**, 30 transfers each, **actual volume 5.0 nL on every one of the 1,164 transfers, zero short transfers**, and the WT source well went 59.8 -> 59.3 uL, i.e. essentially undepleted. The three plates were dispensed 4-5 minutes apart (17:31:52 / 17:36:16 / 17:41:00 on 2026-07-21) from the same source.

5. Not positional bias – and position predicts the opposite sign. The gap through each normalization stage:

plate	raw	+ row/col	+ spatial	final
P1	1.287	1.286	1.275	1.275
P2	0.892	0.973	0.963	0.963
P3	1.325	1.295	1.300	1.300

Row/column correction recovers **+0.081**, so about a quarter of the raw gap IS positional and the pipeline already removes it. The remaining 0.963 versus 1.275 / 1.300 is not. And the sign runs the wrong way: edge colonies grow **larger**, and P2’s wild type is the **most** edge-weighted of the three (37% in the outer two rings, mean edge 2.60, versus 17% and 3.43 on P1). Position predicts P2’s WT should read HIGH; it reads LOW, so the non-positional deficit is if anything *larger* than the residual.

6. A shift, not scatter. Positional noise widens a distribution; a bad denominator translates it.

plate	WT norm median	WT norm IQR
P1	1.246	0.219
P2	0.970	0.203
P3	1.279	0.164

P2’s centre drops ~0.28 while its spread is unchanged.

7. Not a local agar defect or dry region. The low half of P2’s WT wells spans **10 of 16 rows** (9/16 on P1 and P3) – the deficit is spread across the whole plate, not confined to a patch.

8. Not a detection artifact or a biased surviving subset. All **30 of 30** wild-type wells were detected and unflagged on every plate, so the WT median is computed from the full set.

9. Not a normalization gap. Residual positional signal after row/col + spatial correction is $|\rho| \leq 0.037$ against distance from the plate edge (74-96% of the raw bias removed), and median **norm** is flat across edge rings (0.984-1.015).

The downstream signature is internally consistent

If the denominator is depressed to roughly the median mutant, then (a) the median strain should score ~1.0 and (b) about half the strains should exceed WT. Both hold, and only on P2:

plate	strains above WT	median fitness
run3 P1	0 / 12	0.778
run3 P2	6 / 12	1.000
run3 P3	0 / 12	0.771
run2 P2 t44	0 / 12	0.796
run2 P2 t50	0 / 12	0.861

A median of *exactly* 1.000 is the arithmetic fingerprint of a reference equal to the median strain. And 6/12 above wild type is biologically impossible: across 7,738 genome-wide Costanzo deletion strains only **2.7%** are significantly above WT (`ladder_feasibility.py`).

Why our QC let it through

WT_CV = 0.144 on P2 versus 0.066 and 0.079 – clearly elevated, but under the 0.18 gate, so the plate passed. Two sharper gates are now implemented in `between_day_variance.py`:

1. **WT / mutant median size ratio in raw pixels** – 1.17-1.33 on four sound plates, 0.892 on P2. Normalization-independent, and it names the mechanism.
2. **Fraction of deletion strains scoring above WT** – 0/12 on four plates, 6/12 on P2, against a genome-wide expectation of 2.7%.

Gate 1 is the cause, gate 2 the symptom. Both should move into the main scoring path.

Observed vs inferred vs unknown

- **Observed:** P2's wild-type colonies are ~25% smaller relative to that plate's own mutants than on any other plate, after full positional correction, uniformly across the plate, with the mutants unchanged and delivery verified identical.
- **Inferred:** something specific to the BY4741 wells on that one plate, arising *after* dispense – every instrument and software explanation is excluded.
- **Unknown:** whether that was a handling event or genuine variation in that wild-type aliquot. One anomalous plate out of five cannot distinguish them, and the earlier “WT source well” guess was **withdrawn** when the Echo record contradicted it.

Consequences

1. **P2's fitness values are unusable** – all of them are divided by a bad denominator. Its *mutant colony sizes* are fine, so the plate is not worthless; only its normalization is.
2. **$\sigma_{\text{plate}} = 0.140$ is not a noise estimate.** Recomputed on the clean plates it is **0.028**, five times tighter. Every SE, power and design number derived from 0.140 in the next-round layout note is therefore built on one plate's failure and needs revisiting – with the caveat that 0.028 rests on two plates (1 degree of freedom).
3. **The noise model is probably not one Gaussian** but “clean plates agree to ~0.03, and plates occasionally fail outright” (1 of 3 here). That changes the design goal from averaging noise down to **detecting failures and carrying enough plates to survive discarding one.**

Figures

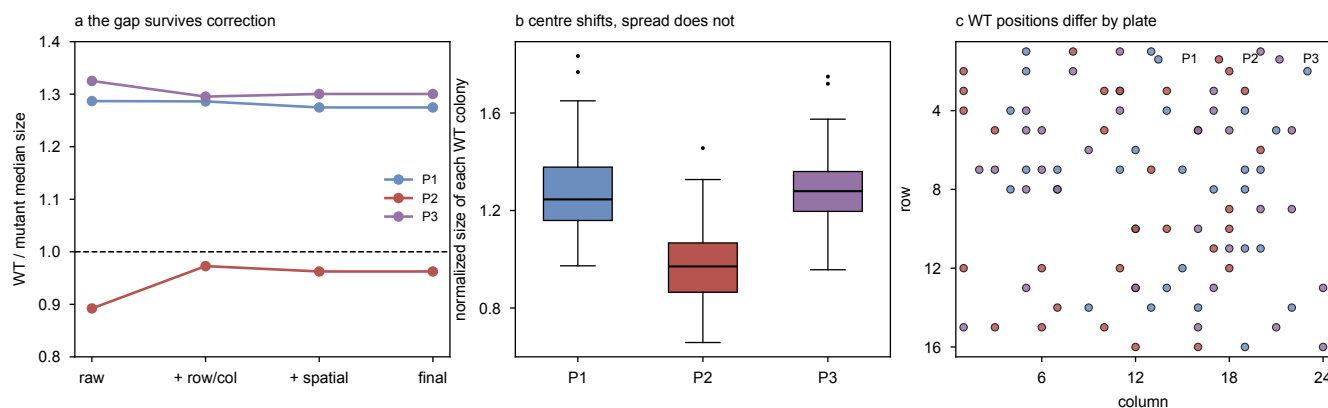


Figure 1: Is the P2 wild-type anomaly positional?

Related notes

- `[[experiments.019-echo-crispr-array.scripts.next_round_layout]]` – design consequences
- `[[experiments.019-echo-crispr-array.scripts.ladder_feasibility]]` – the 2.7% genome-wide figure
- `[[experiments.019-echo-crispr-array.next-round-sampling-design]]` – power calculations

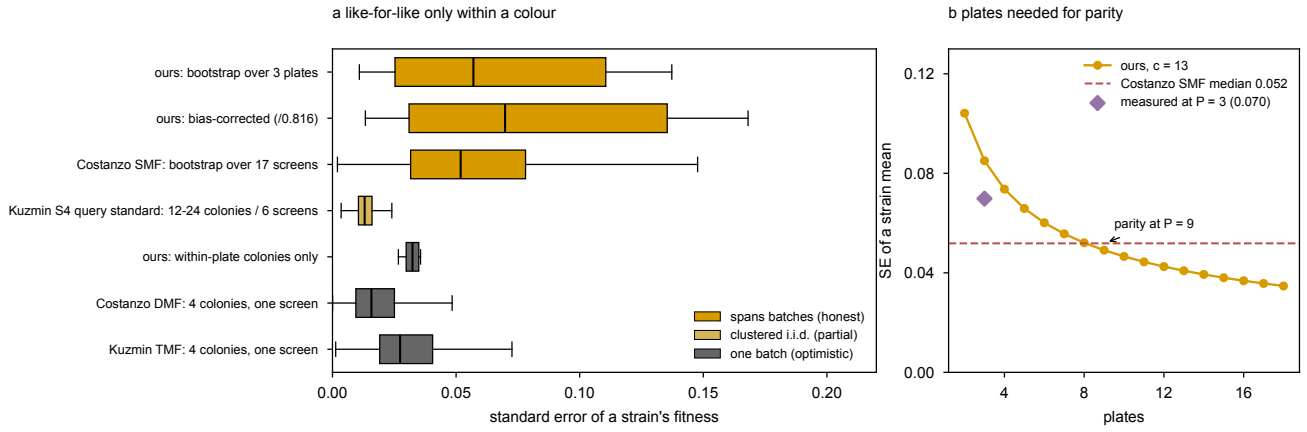


Figure 2: Assay precision against the published SGA screens